Chapter 7: Bayesian Estimation

Part I — Statistics Foundation for ML

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Abstract

Maximum likelihood (Chapter 6) treats the parameter as a fixed unknown and asks which value best explains the data. The Bayesian view instead treats the parameter as a random quantity with a distribution: you start with a *prior*, see data, and update to a *posterior* by Bayes' rule. This chapter builds that update, shows the conjugate families where it stays a one-line formula, and closes by connecting the two philosophies — with enough data the posterior concentrates into the same normal the maximum-likelihood estimator obeys, so Bayesian and frequentist answers meet.

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1 Prior, likelihood, posterior

Write $\pi(\theta)$ for the **prior** — what you believe about the parameter before seeing data — and $L(x \mid \theta)$ for the likelihood. Bayes' rule combines them into the **posterior**.

Definition 1.1 (Posterior)**.**

$$\pi(\theta \mid x) = \frac{L(x \mid \theta)\,\pi(\theta)}{\int L(x \mid \theta')\,\pi(\theta')\,d\theta'} \;\propto\; L(x \mid \theta)\,\pi(\theta).$$

The denominator is a constant in θ, so the shape of the posterior is just likelihood times prior; it is normalised at the end to integrate to one.

From the posterior come the usual summaries: the **posterior mean** $E[\theta \mid x]$, the **maximum a posteriori** (MAP) estimate $\arg\max_\theta \pi(\theta \mid x)$, and a **credible interval**, an interval holding, say, 95% of the posterior probability.

The cleanest case is when prior and posterior belong to the same family — the prior is **conjugate** to the likelihood — so updating only changes the parameters. The headline example is the Beta prior with Bernoulli/Binomial data.

Theorem 1.1 (Beta–Binomial conjugacy)**.** With prior $\theta \sim \text{Beta}(\alpha, \beta)$ and k successes in n independent trials, the posterior is $\theta \mid x \sim \text{Beta}(\alpha + k,\ \beta + n - k)$.

Its posterior mean, $\frac{\alpha+k}{\alpha+\beta+n}$, is a blend of the prior and the data that slides toward the sample proportion k/n as n grows. Conjugacy makes several models one-liners:

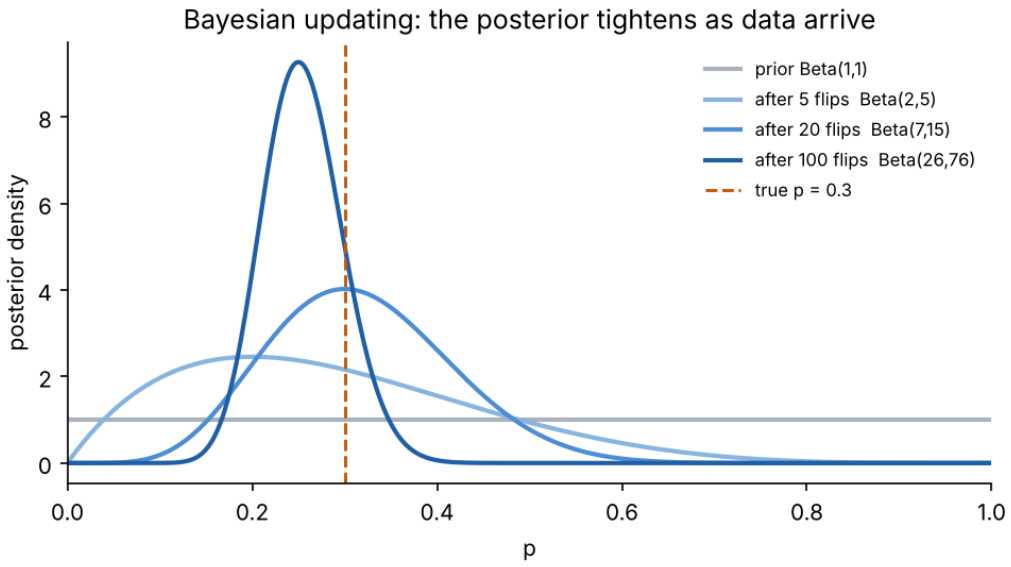
```python
def beta_binomial(alpha, beta, k, n):                # Beta prior, Binomial data
    return (alpha + k, beta + n - k)                 # -> Beta(.,.)

def gamma_poisson(a, b, total, n):                   # Gamma(a,b) prior, Poisson
    data
    return (a + total, b + n)                        # -> Gamma(shape, rate)

def normal_normal(mu0, tau2, xbar, sigma2, n):       # Normal prior, known
    variance
    prec = 1 / tau2 + n / sigma2
    mean = (mu0 / tau2 + n * xbar / sigma2) / prec
    return (mean, 1 / prec)                          # -> Normal(mean, variance)

print(beta_binomial(1, 1, 8, 10))                    # (9, 3)
print(gamma_poisson(2, 1, 30, 10))                   # (32, 11)
print(normal_normal(0, 1, 2.0, 1.0, 10))             # (1.818..., 0.0909...)
```

Each line is a complete inference: feed the prior parameters and the data, read off the posterior. The figure shows the Beta–Binomial case as flips accumulate — a flat prior sharpening into a tight posterior around the true p.



2 Why the updates work

Both conjugate formulas above come straight from "posterior $\propto$ likelihood $\times$ prior," with the algebra being a matter of collecting exponents.

Theorem 2.1 (Proof of Beta–Binomial)**.** The Binomial likelihood in θ is $\theta^k(1-\theta)^{n-k}$ and the Beta prior is $\propto \theta^{\alpha-1}(1-\theta)^{\beta-1}$. Multiplying,

$$\pi(\theta \mid x) \;\propto\; \theta^k(1-\theta)^{n-k}\,\theta^{\alpha-1}(1-\theta)^{\beta-1} = \theta^{(\alpha+k)-1}(1-\theta)^{(\beta+n-k)-1},$$

which is the kernel of $\mathrm{Beta}(\alpha+k,\ \beta+n-k)$.

The Normal case is just as mechanical, and the answer is worth stating in words: posterior precisions add, and the posterior mean is a precision-weighted average.

Theorem 2.2 (Normal–Normal, known variance)**.** With prior $\theta \sim N(\mu_0, \tau^2)$ and n observations of mean $\bar{x}$ from $N(\theta, \sigma^2)$, the posterior is normal with

$$\text{precision } \frac{1}{\tau^2} + \frac{n}{\sigma^2}, \qquad \text{mean } \frac{\mu_0/\tau^2 + n\bar{x}/\sigma^2}{1/\tau^2 + n/\sigma^2}.$$

Sketch. Both prior and likelihood are exponentials of quadratics in θ. Adding the exponents gives another quadratic, $-\frac{1}{2}\big(\frac{1}{\tau^2} + \frac{n}{\sigma^2}\big)\theta^2 + \big(\frac{\mu_0}{\tau^2} + \frac{n\bar{x}}{\sigma^2}\big)\theta + \text{const}$. Completing the square reads off a normal whose precision is the coefficient of $-\frac{1}{2}\theta^2$ and whose mean is the stated ratio. □

Notice what the weights do: a tight prior (small τ^2) or a lot of data (large n) pulls the posterior mean toward that side. The prior behaves like $1/\tau^2$ worth of "pseudo-observations."

3 Priors, prediction, and the link to maximum likelihood

Two estimates summarise a posterior, and they usually differ: the **posterior mean** $E[\theta \mid x]$ minimises expected squared error, while the **MAP** estimate $\arg\max \pi(\theta \mid x)$ is the posterior mode. For a symmetric posterior they coincide; for a skewed one they do not. MAP with a flat prior is just maximum likelihood, which is the first hint that the two frameworks are close.

To predict a new observation one averages the likelihood over the posterior rather than plugging in a point estimate — the **posterior predictive**

$$p(x_{\text{new}} \mid x) = \int p(x_{\text{new}} \mid \theta)\, \pi(\theta \mid x)\, d\theta,$$

which carries the parameter uncertainty forward instead of pretending θ is known exactly.

Where does the prior come from? One principled default is the **Jeffreys prior** $\pi(\theta) \propto \sqrt{I(\theta)}$, built from the Fisher information of Chapter 6 so that it is invariant to how the parameter is written. And whatever the prior, its influence fades: as data accumulate, the likelihood dominates and the posterior forgets where it started. The precise statement is the bridge back to Chapter 6.

Theorem 3.1 (Bernstein–von Mises, informal)**.** Under regularity, for large n the posterior is approximately

$$\theta \mid x \;\approx\; N\big(\hat{\theta}_{\text{MLE}},\; I_n(\theta)^{-1}\big),$$

centred at the maximum-likelihood estimate with variance the inverse Fisher information.

So in large samples a Bayesian credible interval and a frequentist confidence interval very nearly coincide, and the choice of prior stops mattering. The approximation itself — fit a normal at the posterior mode using the local curvature — is the **Laplace approximation**, the workhorse behind practical Bayesian deep learning in Part III.

Summary

Bayesian estimation updates a prior to a posterior by $\pi(\theta \mid x) \propto L(x \mid \theta)\pi(\theta)$. Conjugate families keep this a formula: Beta–Binomial gives $\text{Beta}(\alpha + k, \beta + n - k)$, Normal–Normal adds precisions and takes a precision-weighted mean. Summaries are the posterior mean, the MAP estimate, and credible intervals; prediction integrates over the posterior. As $n \to \infty$ the posterior concentrates to $N(\hat{\theta}_{\text{MLE}}, I_n^{-1})$ (Bernstein–von Mises), so the Bayesian and maximum-likelihood answers meet.